

SIMATS ENGINEERING
ASSESSMENT TOOL 3: MCQ Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 10%

Dave's Taxonomy: Precision (Level 3)
Question Count: 15

Total Marks:
30

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student: _____

Assessment Tasks

MCQ Test

1. Which cloud storage model is best suited for storing virtual machine disks?
 - A. Object storage
 - B. Block storage
 - C. File storage
 - D. Archive storage
2. Which protocol style is widely used for lightweight web APIs in cloud platforms?
 - A. SOAP only
 - B. REST
 - C. FTP
 - D. SMTP
3. Serverless computing is best described as:
 - A. Computing without networks
 - B. Running code without managing servers directly
 - C. Local execution only
 - D. Batch-only processing
4. Edge computing primarily helps by:
 - A. Increasing central storage only
 - B. Reducing latency near data sources
 - C. Eliminating internet access
 - D. Replacing cloud storage fully
5. Fog computing differs from edge computing because it:
 - A. Always runs inside sensors

- B. Introduces an intermediate distributed layer between edge and cloud
 - A. Works only in private clouds
 - B. Uses no network resources
6. Which storage model is commonly used for shared departmental files?
- A. File storage
 - B. Register storage
 - C. Cache-only storage
 - D. Tape-only storage
7. A browser calling a cloud web application typically communicates through:
- A. BIOS interrupts
 - B. Web APIs
 - C. Only command-line tools
 - D. VPN tunnels only
8. SAN stands for:
- A. Secure Access Node
 - B. Storage Area Network
 - C. Shared Application Namespace
 - D. Service Access Network
9. Which of the following is a common benefit of serverless architecture?
- A. Manual scaling of servers
 - B. Pay-per-use execution model
 - C. Dedicated physical hardware ownership
 - D. No monitoring required
10. Collaboration within the cloud usually improves:
- A. Shared access to services and documents
 - B. Hardware disposal speed
 - C. Cable management
 - D. BIOS configuration
11. Which one is an example of a platform service?
- A. Managed application runtime
 - B. Keyboard driver
 - C. Physical rack switch only
 - D. Local USB storage
12. Object storage is generally preferred for:
- A. Mounting OS boot volumes
 - B. Large unstructured cloud data such as images and backups
 - C. CPU scheduling
 - D. Hypervisor installation
13. Which component allows applications to communicate programmatically with cloud services?
- A. Heat sink
 - B. Web API
 - C. Mouse driver
 - D. Compiler cache
14. A low-latency smart factory use case is best supported by:
- A. Edge computing
 - B. Cold archive only
 - C. Offline storage
 - D. Manual processing
15. Which one is not a typical cloud access method?
- A. Browser-based access
 - B. API access

- C. Platform console access
- D. Paper-based access

16. Which storage model is commonly used for shared departmental files?

- A. File storage
- B. Register storage
- C. Cache-only storage
- D. Tape-only storage

17. A browser calling a cloud web application typically communicates through:

- A. BIOS interrupts
- B. Web APIs
- C. Only command-line tools
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- A. Secure Access Node
- B. Storage Area Network
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19. Which of the following is a common benefit of serverless architecture?

- A. Manual scaling of servers
- B. Pay-per-use execution model
- C. Dedicated physical hardware ownership
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21. Which one is an example of a platform service?

- A. Managed application runtime
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22. Object storage is generally preferred for:

- A. Mounting OS boot volumes
- B. Large unstructured cloud data such as images and backups
- C. CPU scheduling
- D. Hypervisor installation

23. Which component allows applications to communicate programmatically with cloud services?

- A. Heat sink
- B. Web API
- C. Mouse driver
- D. Compiler cache

24. A low-latency smart factory use case is best supported by:

- A. Edge computing
- B. Cold archive only
- C. Offline storage
- D. Manual processing

25. Which one is not a typical cloud access method?

- A. Browser-based access
- B. API access
- C. Platform console access

D. Paper-based access

MCQ Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Understanding	30%	Answers reflect strong and accurate concept mastery	Mostly accurate understanding	Partial understanding	Weak understanding
Accuracy of Responses	30%	Nearly all responses are correct	Most responses are correct	Some responses are correct	Few responses are correct
Application Awareness	15%	Shows strong recognition of practical cloud use cases	Reasonable recognition	Limited recognition	Weak recognition
Time Management	10%	Completes assessment efficiently	Completes on time	Needs extra time	Unable to finish well
Question Interpretation	15%	Interprets all items correctly	Misreads very few items	Misreads several items	Frequently misinterprets questions
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning

1. Which cloud storage model is best suited for storing virtual machine disks?

- A. Object storage
- B. Block storage
- C. File storage
- D. Archive storage

Answer: B. Block storage

Explanation: Block storage provides block-level volumes that can be attached to virtual machines and used like disks.

2. Which protocol style is widely used for lightweight web APIs in cloud platforms?

- A. SOAP only
- B. REST
- C. FTP
- D. SMTP

Answer: B. REST

Explanation: REST is widely used for lightweight web APIs because it uses standard HTTP methods for communication between applications and cloud services.

3. Serverless computing is best described as:

- A. Computing without networks
- B. Running code without managing servers directly
- C. Local execution only
- D. Batch-only processing

Answer: B. Running code without managing servers directly

Explanation: Serverless computing allows developers to run application code without directly managing the underlying servers.

4. Edge computing primarily helps by:

- A. Increasing central storage only
- B. Reducing latency near data sources
- C. Eliminating internet access
- D. Replacing cloud storage fully

Answer: B. Reducing latency near data sources

Explanation: Edge computing processes data close to where it is generated, reducing the time required to send data to a central cloud.

5. Fog computing differs from edge computing because it:

- A. Always runs inside sensors
- B. Introduces an intermediate distributed layer between edge and cloud
- C. Works only in private clouds
- D. Uses no network resources

Answer: B. Introduces an intermediate distributed layer between edge and cloud

Explanation: Fog computing provides an intermediate distributed computing layer between edge devices and the central cloud.

6. Which storage model is commonly used for shared departmental files?

- A. File storage
- B. Register storage
- C. Cache-only storage
- D. Tape-only storage

Answer: A. File storage

Explanation: File storage provides shared folders and files that can be accessed by multiple users and applications.

7. A browser calling a cloud web application typically communicates through:

- A. BIOS interrupts
- B. Web APIs
- C. Only command-line tools
- D. VPN tunnels only

Answer: B. Web APIs

Explanation: Web APIs allow browsers and other applications to communicate programmatically with cloud-based web applications.

8. SAN stands for:

- A. Secure Access Node
- B. Storage Area Network
- C. Shared Application Namespace
- D. Service Access Network

Answer: B. Storage Area Network

Explanation: SAN stands for Storage Area Network. It is a specialized network that provides access to block-level storage.

9. Which of the following is a common benefit of serverless architecture?

- A. Manual scaling of servers
- B. Pay-per-use execution model
- C. Dedicated physical hardware ownership
- D. No monitoring required

Answer: B. Pay-per-use execution model

Explanation: Serverless platforms commonly charge based on actual execution or resource usage, reducing the need to maintain continuously running servers.

10. Collaboration within the cloud usually improves:

- A. Shared access to services and documents
- B. Hardware disposal speed
- C. Cable management
- D. BIOS configuration

Answer: A. Shared access to services and documents

Explanation: Cloud collaboration allows multiple users to access, share, and work on services and documents from different locations.

11. Which one is an example of a platform service?

- A. Managed application runtime
- B. Keyboard driver
- C. Physical rack switch only
- D. Local USB storage

Answer: A. Managed application runtime

Explanation: A managed application runtime is a platform service that provides an environment for developing and running applications without managing the underlying infrastructure directly.

12. Object storage is generally preferred for:

- A. Mounting OS boot volumes
- B. Large unstructured cloud data such as images and backups

- C. CPU scheduling
- D. Hypervisor installation

Answer: B. Large unstructured cloud data such as images and backups

Explanation: Object storage is designed for large amounts of unstructured data such as images, videos, documents, and backups.

13. Which component allows applications to communicate programmatically with cloud services?

- A. Heat sink
- B. Web API
- C. Mouse driver
- D. Compiler cache

Answer: B. Web API

Explanation: A Web API provides a programmatic interface that allows applications to communicate with cloud services and exchange data.

14. A low-latency smart factory use case is best supported by:

- A. Edge computing
- B. Cold archive only
- C. Offline storage
- D. Manual processing

Answer: A. Edge computing

Explanation: Edge computing processes data close to machines and sensors, making it suitable for smart factories that require fast, low-latency decisions.

15. Which one is not a typical cloud access method?

- A. Browser-based access
- B. API access
- C. Platform console access
- D. Paper-based access

Answer: D. Paper-based access

Explanation: Cloud services are normally accessed through browsers, APIs, or cloud platform consoles. Paper-based access is not a cloud access method.

16. Which storage model is commonly used for shared departmental files?

- A. File storage
- B. Register storage
- C. Cache-only storage
- D. Tape-only storage

Answer: A. File storage

Explanation: File storage is commonly used for shared departmental files because multiple users can access files and folders through a shared file system.

17. A browser calling a cloud web application typically communicates through:

- A. BIOS interrupts
- B. Web APIs
- C. Only command-line tools
- D. VPN tunnels only

Answer: B. Web APIs

Explanation: Web APIs allow a browser or client application to communicate with a cloud web application and exchange data.

18. SAN stands for:

- A. Secure Access Node
- B. Storage Area Network
- C. Shared Application Namespace
- D. Service Access Network

Answer: B. Storage Area Network

Explanation: SAN stands for Storage Area Network. It is a dedicated network that provides access to storage resources.

19. Which of the following is a common benefit of serverless architecture?

- A. Manual scaling of servers
- B. Pay-per-use execution model
- C. Dedicated physical hardware ownership
- D. No monitoring required

Answer: B. Pay-per-use execution model

Explanation: Serverless architecture commonly follows a pay-per-use model, where users are charged based on the resources or execution time they consume.

20. Collaboration within the cloud usually improves:

- A. Shared access to services and documents
- B. Hardware disposal speed
- C. Cable management
- D. BIOS configuration

Answer: A. Shared access to services and documents

Explanation: Cloud collaboration allows multiple users to access and work with shared services and documents from different locations.

21. Which one is an example of a platform service?

- A. Managed application runtime
- B. Keyboard driver
- C. Physical rack switch only
- D. Local USB storage

Answer: A. Managed application runtime

Explanation: A managed application runtime is a platform service that provides an environment for developing and running applications without directly managing the underlying infrastructure.

22. Object storage is generally preferred for:

- A. Mounting OS boot volumes
- B. Large unstructured cloud data such as images and backups
- C. CPU scheduling
- D. Hypervisor installation

Answer: B. Large unstructured cloud data such as images and backups

Explanation: Object storage is suitable for large unstructured data such as images, videos, documents, and backup files.

23. Which component allows applications to communicate programmatically with cloud services?

- A. Heat sink
- B. Web API
- C. Mouse driver
- D. Compiler cache

Answer: B. Web API

Explanation: A Web API provides an interface that allows applications to communicate programmatically with

cloud services.

24. A low-latency smart factory use case is best supported by:

- A. Edge computing
- B. Cold archive only
- C. Offline storage
- D. Manual processing

Answer: A. Edge computing

Explanation: Edge computing processes data close to the sensors and machines, reducing latency and enabling quick decisions in smart factory environments.

25. Which one is not a typical cloud access method?

- A. Browser-based access
- B. API access
- C. Platform console access
- D. Paper-based access

Answer: D. Paper-based access

Explanation: Browser-based access, API access, and platform console access are common ways to access cloud services. Paper-based access is not a cloud access method.